

# Abhirup Paul

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## SUMMARY

Backend software engineer building well-architected, scalable enterprise-grade solutions in **Golang**, Python, and C++ across high-traffic, real-time systems — data analytics services, payment orchestration, and event-driven microservices. Strong with software design patterns, Git, Jira, Docker/Kubernetes, SQL databases, Make, and CI/CD (GitLab CI, GitHub Actions); cross-platform development on Linux and Windows. Co-inventor of a patented clinical platform (Neuroscan360) deployed in top-tier hospitals.

## EDUCATION

**GKV FET, Haridwar**

*B.Tech, Electronics and Computer Engineering — CGPA: 8.5 / 10.0*

Uttarakhand, India

2022 – 2026

## EXPERIENCE

**R&D Engineer — SCAN Lab, IIT Bombay**

*Indian Institute of Technology Bombay*

Mumbai, India

Jul 2025 – Present

- Architected **Neuroscan360**, a patented full-stack clinical screening platform for neurological assessments, live in Lilavati Hospital and Kokilaben Dhirubhai Ambani Hospital; used daily by multiple medical teams.
- Built scalable Python server applications (FastAPI) and a **Golang** data-analytics service for batch processing of multimodal patient records; containerised all services with **Docker**, orchestrated on **Kubernetes (k8s)**, and shipped via CI/CD (GitLab CI, GitHub Actions) with cross-platform support across Linux and Windows.
- Built the cross-platform React Native mobile app (iOS & Android) from scratch — device sensor integration, real-time multimodal data capture, and clinical workflow UX; pending App Store & Play Store publication.
- Designed FastAPI microservices with **sub-50ms inference latency**, serving real-time ML models for neurological assessment over REST and WebSocket.
- Owned end-to-end PostgreSQL (SQL) schema for multimodal patient data; deployed on AWS Well-Architected Framework — improved system reliability by **64%** and cut infrastructure cost by **30%**.

**Backend Engineer (Golang) — Contract**

*Independent*

Remote

Feb 2025 – Jul 2025

- Built scalable, enterprise-grade backend microservices in **Golang**, handling concurrent high-throughput request loads using goroutines and channels; designed clean, idiomatic Go applying established software design patterns.
- Developed and consumed REST APIs backed by **SQL (PostgreSQL)** databases; containerised services with **Docker** and deployed on **Kubernetes**, with cross-platform builds for both Linux and Windows.
- Managed source control with **Git**, tracked work in **Jira**, automated builds using **Make**, and set up Continuous Integration via **GitLab CI**; communicated technical trade-offs clearly to stakeholders. *Stack: Go, PostgreSQL, Docker, Kubernetes, GitLab CI*

**Freelance Backend Engineer**

*Independent*

Remote

Dec 2024 – Feb 2025

- Designed and shipped a rule-based **fraud signal engine** processing real-time financial transactions — full ownership of API design, data modelling, event processing, and infrastructure.
- Built a WebSocket-based real-time alerting system with ledger-aware backend ensuring transaction integrity and preventing duplicate/replayed events; secured APIs with JWT authentication.
- PostgreSQL schema + Redis caching layer for low-latency lookups; chose Redis Pub/Sub over managed queue to reduce latency and operational overhead. *Stack: Python, FastAPI, PostgreSQL, Redis, AWS*

## PROJECTS

**Distributed Data-Analytics Pipeline (Golang) | Go, PostgreSQL, Docker, Kubernetes, GitLab CI**

- Built a high-throughput, scalable analytics service in **Golang** ingesting and aggregating large event datasets concurrently with goroutines and worker pools; exposed results over a clean REST API backed by **SQL (PostgreSQL)**.
- Applied software design patterns (repository, factory, dependency injection) for testable, enterprise-grade code; containerised with **Docker**, deployed on **Kubernetes**, automated builds via **Make** and **GitLab CI**, with cross-platform support on Linux and Windows.

## TECHNICAL SKILLS

**Languages:** Golang (Go), Python, C++, Java, JavaScript, TypeScript, SQL

**Backend:** REST API design, Microservices, Event-Driven Architecture, Concurrency (goroutines/channels), WebSocket, Software Design Patterns, JWT

**Databases:** PostgreSQL (pgvector), MongoDB, Redis, Firebase, SQL Databases, Vector Databases

**DevOps & Platforms:** Docker, Kubernetes (k8s), GitLab CI, GitHub Actions, CI/CD, Git, Jira, Make, AWS (Well-Architected), Cloud Providers, Cross-platform (Linux & Windows), Distributed Systems

**AI / LLM:** LangChain, RAG Pipelines, LLM API Integration, Claude API, OpenAI API; AI-Native Dev Tools (Claude Code, Cursor, GitHub Copilot)

**Mobile & Frontend:** React Native (iOS & Android), React.js, Next.js, Redux

## ACHIEVEMENTS

Co-inventor — **Neuroscan360** patent filed at IIT Bombay; live in Lilavati & Kokilaben hospitals; reduced AWS cost **30%** and improved reliability **64%** via deliberate architecture

Runner-up — **MumbaiHacks 2024**; shipped production systems used by hospital clinicians and financial-platform clients